

Difference b/w volatile and Non-Volatile Memory

Volatile Memory:

- Requires Power To maintain the stored information or Data.
- Temporary memory.
Examples: (primary)
 - Random Access Memory
 - Cache.
 - Register memory.

Non-Volatile Memory:

- It doesn't require power to Retain information or Data.
- Permanent memory.
Examples:
 - Flash memory
 - Magnetic storage
 - Optical storage
 - Read only memory (ROM) → (primary)

Volatility

vs

Non-Volatility

- Requires a constant power supply.

- Stores its content temporarily.

- Very Fast.

- It is a primary storage.

- Smaller storage capacity.

- More expensive per unit size.

- Will retain its contents even with no power supply.

- Stores its contents permanently.

- Relatively Slow.

- It is a secondary storage.

- Larger storage capacity.

- Less expensive per unit size.

Unit size: per unit size:

: Questions about Volatility :

Q1.: What is volatility in the context of computer memory?

Ans.: Volatility refers to the loss of Data when power is turned off.

Q2.: How does volatility effect data storage in computers?

Ans.: Volatile memory loses data when power is turned off, so data must be saved to non-volatile storage to be retained.

Q3.: What are the advantages and disadvantages of volatile memory?

Ans.: Advantages: Fast access times, low cost.
Disadvantages: Loses data when power is turned off.

Q4.: Can you give ^{up} volatile memory technologies?

Ans.: Example include RAM (Random access memory) and cache memory.

Q5.: How does volatility impact the performance of a computer system?

Ans.: Volatility can impact performance by requiring data to be reloaded from non-volatile storage when power is turned off on.

: Questions about Non-Volatility :

Q1.: What is non volatility in the context of computer memory?

Ans.: Non-volatility refers to the retention (retain) of data even when power is turned off.

Q2: How does non-volatility differ from volatility in terms of data storage?

Ans: Non-volatile memory retains data even when power is turned off, while volatile memory loses data.

Q3: What are advantages and disadvantages of non-volatile-memory?

Advantages: Retains data even when power is turned off, reliable.

Disadvantages: slower access times, higher cost.

Q4: Can you give 3 examples of non-volatile memory technologies?

Ans: Examples include hard disk drives, solid-state drives, and flash memory.

Q5: How does non-volatility improve ability impact the performance and reliability of a computer system?

Ans: Non-volatility improves reliability by retaining data even power is turned off, but may impact performance due to slower access time.